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White Paper

Application of the European Union's Artificial Intelligence Act 2024/1689 to the medical device industry:

Opportunities and challenges

Abstract

The Artificial Intelligence (AI) Act is the world's first comprehensive legal framework for AI. It aims to foster trustworthy AI in Europe through a clear set of risk-based rules for developers and deployers. This white paper examines how the AI Act applies to medical devices and outlines the obligations it places on manufacturers, providers, and users. It explores the current landscape of knowns and unknowns, and the AI Act's interaction with the Medical Device Regulation (MDR). Finally, it reviews the state of conformity assessments for medical AI devices and how these can support a strong foundation for meeting AI Act requirements.

Contents

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Introduction

The AI Act is the first comprehensive legislative framework for regulating AI within the European Union (EU). First proposed in April 2021 by the European Commission, the AI Act has been years in the making. Building upon previous EU initiatives, including the 2018 Coordinated Plan on AI[1] and the 2020 White Paper on AI[2], the Act represents a significant legislative effort. The European Council issued its position in December 2022, and the European Parliament adopted its negotiating position in June 2023.

The final version of the AI Act was endorsed by all 27 member states in February 2024. Published in the EU's Official Journal on 12 July 2024, the Act became binding law from 1 August 2024, irrespective of existing national laws and guidelines[3].

Providers of high-risk AI systems are encouraged to start complying with the Act voluntarily during the transitional period[4]. However, the law applies from 2 August 2026[5], although obligations for high-risk AI systems apply one year later, on 2 August 2027.

The broad scope and stringent requirements of the AI Act have significant implications for the medical device industry. As the sector becomes increasingly reliant on AI for innovation and efficiency, manufacturers, notified bodies and healthcare organisations must be well prepared. Figure A highlights the many individuals who will be impacted by the Act when bringing medical devices that incorporate artificial intelligence to market.

Fig. A Examples (includes but not limited to) of individuals in the medical device industry impacted by the AI Act

Why the AI Act is important

The AI Act is unprecedented in scope. Unlike the MDR, it does not explicitly cover all medical AI applications, and not all AI applications used by the healthcare sector fall within the scope of the MDR, e.g., general-purpose large language models such as ChatGPT.

According to Article 1(1) of the Act, its purpose is to:
- Improve the internal market's functionality.
- Promote human-centric and trustworthy AI.
- Ensure high levels of protection for health, safety, fundamental rights, democracy and the environment.

Definitions
It is important that the medical device industry has a full understanding of the new terms introduced by the AI Act. These are defined in Article 3 of the Act and a summary is provided below in Table 1.

The AI Act contains 113 articles and 13 annexes. Its wide scope incorporates:

- All providers of AI systems in the EU market, regardless of where they are established or located[6].
- *Providers and deployers of AI systems in third countries, whose generated output is used within the EU[7].
- Deployers of AI systems whose place of establishment is in the EU or are located within the EU8.

*Output is not defined in the Act and examples provided are vague

Fig. B Size and scope of the AI Act

Term	Definition	Source	Example
GPAI model	"[...] an AI model, including where such an AI model is trained with a large amount of data using self-supervision at scale, that displays significant generality and is capable of competently performing a wide range of distinct tasks regardless of the way the model is placed on the market and that can be integrated into a variety of downstream systems or applications [...]"	Article 3(63)	GPT-4
GPAI system	"[...] an AI system which is based on a general-purpose AI model, that has the capability to serve a variety of purposes, both for direct use as well as for integration in other AI systems [...]"	Article 3(66)	ChatGPT
AI system	"[...] a machine-based system designed to operate with varying levels of autonomy, that may exhibit adaptiveness after deployment and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments [...]"	Article 3(1)	Facial recognition system
Provider	"[...] means a natural or legal person, public authority, agency or other body that develops an AI system or a general-purpose AI model or that has an AI system or a general-purpose AI model developed and places it on the market or puts the AI system into service under its own name or trademark, whether for payment or free of charge [...]"	Article 3(3)	OpenAI
Deployer	"[...] means a natural or legal person, public authority, agency or other body using an AI system under its authority except where the AI system is used in the course of a personal non-professional activity [...]"	Article 3(4)	A healthcare organisation

Tab. 1 Some key definitions from the EU AI Act

Key aspects of the AI Act relevant to medical devices

Prohibited AI practices

Article 5 of the AI Act outlines practices that are considered prohibited due to unacceptable risks. This includes:

- Purposefully manipulative and deceptive practices.
- Exploitation of vulnerabilities.
- Biometric categorisation.
- Social scoring.
- Real-time remote biometric identification.
- Risk assessments for criminal offences.
- Facial recognition and emotion inference.

The AI Act defines fines for non-compliances of up to 35 million euros or 7% of annual turnover, whichever is higher[9]. However, there are exceptions for medical uses under applicable laws and standards[10]. The exceptions include psychological treatment for mental illness or physical rehabilitation, when carried out following applicable laws and medical standards and with the explicit consent of individuals or their legal representatives[11], as well as facial recognition and emotion recognition systems that are permitted for medical reasons[12].

Risk-based categorisation

The AI Act follows a risk-based approach, similar to the MDR. The Act categorises AI systems into four risk levels: minimal risk, limited risk, high risk and prohibited.

Medical devices incorporating AI are typically classified as high-risk systems due to their direct impact on health and safety. They are classified as high risk if they:

1. Serve as a safety component of a product or are themselves a product covered by EU harmonization laws (such as the MDR) requiring third-party conformity assessment[13]. Thus, all medical AI products classified as MDR risk class IIa or higher must also fulfil the requirements of the AI Act for high-risk systems. An example could be AI-assisted medical image diagnosis software.
2. Are listed in Annex III of the AI Act and pose significant risks to health, safety or fundamental rights of natural persons[14]. Examples include systems intended for emotion recognition (e.g., used to monitor psychiatric treatments) and emergency patient triage systems.

Per Article 6(3) of the AI Act, this type of AI system could be exempt from the requirements of high-risk devices if the device is intended to:

- Perform a narrow procedural task.
- Improve the outcome of a previously performed human activity.
- Detect decision patterns or deviations from previous decision patterns and is not intended to replace or influence the previously performed human assessment without proper human review.

or

- Perform a preparatory task for an assessment relevant to the purpose of the use cases listed in Annex III.

Therefore, common AI system administrative tasks in the medical field, such as medical text classification (e.g., ICD-10 coding) or structuring (e.g., structured radiology reporting) are unlikely to be classified as high risk, unless falling under point 1 above. It is specified within the AI Act that the EU Commission will provide a more comprehensive list of practical examples of high-risk and non-high risk use cases within 18 months after it enters into force[15]. Figure C highlights strict requirements that apply to high-risk AI systems.

Fig. C High-risk AI systems requirements

General purpose artificial intelligence (GPAI) models

GPAI models can generally be used as standalone high-risk AI systems or as components of other AI systems within any risk class. GPAI models must always fulfil certain requirements as their capabilities allow for several downstream tasks[16].

GPAI models are classified according to whether they present systemic risks or not. In Article 3(65), a systemic risk is defined as: "a risk that is specific to the high-impact capabilities of general-purpose AI models, having a significant impact on the Union market due to their reach, or due to actual or reasonably foreseeable negative effects on public health, safety, public security, fundamental rights, or the society as a whole, that can be propagated at scale across the value chain." This distinction is important because GPAI models with systemic risks have additional compliance requirements, including those related to model evaluation, risk mitigation and management, and cybersecurity[17].

Supervision distinctions between AI systems and GPAI models

The distinction between AI systems and GPAI models and systems is important from the perspective of regulatory oversight. The AI Act stipulates that AI systems should be supervised by national market surveillance authorities, whereas GPAI models and systems should be supervised by a newly established AI Office at the EU level[18].

Transparency obligations

Irrespective of the risk classification attributed to an AI model or system, every one that interacts with natural persons or generates content such as text, images or videos must fulfil additional transparency obligations. In particular:

- Users must be informed about the use of AI[19].
- Outputs must be machine-readable and recognisable as artificially generated or manipulated[20].

Examples of AI models that must meet these transparency obligations, but may not be considered high-risk, include virtual health assistants and chatbots.

For high-risk AI systems, the AI Act mandates that these provide information so that users can understand their functionality and limitations[21]. For medical devices, this includes the clear labelling of AI-driven features, documentation of decision-making processes, and user training to mitigate risks associated with AI use.

Exemptions from the AI Act

There are some AI systems that are exempt from the requirements of the AI Act. These comprise of:

- AI systems developed and used solely for the purpose of scientific research or for personal, non-professional activities[22].
- AI systems released under free and open-source licenses, unless they use prohibited practices, are classified as high risk, or are subject to additional transparency obligations if they interact directly with individuals[23]. However, if the open-source systems are monetised (e.g., by providing paid technical support), they are subject to the same rules as their closed-source counterparts.

AI Act and MDR requirement discrepancies: Implications for manufacturers and notified bodies

The introduction of the AI Act alongside the MDR has resulted in a complex regulatory landscape. While they share common goals of safety, transparency and accountability, there are some contradictory requirements. This creates compliance challenges for manufacturers and notified bodies responsible for conformity assessments.

Some of the key contradictory aspects between the AI Act and MDR are as follows:

- **Risk classification:** The MDR categorises medical devices based on clinical risk (class I–III), whereas the AI Act follows a four-tier system (minimal, limited, high-risk, prohibited). This can result in a medical AI device being classified as Class IIa under the MDR, while simultaneously being considered high-risk under the AI Act and having additional compliance requirements.

  Manufacturers must align both the AI Act and MDR classifications of their devices to facilitate consistency in the conformity assessment process across both.

- **The treatment of software updates:** The interpretation of the MDR generally requires re-certification for any 'significant' modifications to medical device software, including changes to any AI model that impacts clinical safety. In contrast, the AI Act allows continuous learning AI models to evolve post-market, provided they remain within pre-defined parameters.

  This raises the possibility of uncertainty for manufacturers and notified bodies in situations when an AI model update is considered minor under the AI Act but substantial under the MDR, which may potentially trigger a new conformity assessment. Clear guidelines must be established to define when AI model modifications necessitate regulatory re-evaluation.

- **Transparency obligations:** The requirements and expectations for transparency are not fully aligned between the AI Act and the MDR. The AI Act mandates explainability and disclosure of AI decision-making logic, to ensure that end-users understand how AI-based systems arrive at specific conclusions. However, the MDR focuses primarily on clinical validation and performance evidence, without requiring manufacturers to reveal proprietary AI algorithms.

  This tension between regulatory transparency requirements and intellectual property protection means that notified bodies must adopt assessment methods that respect trade secrets while ensuring compliance with explainability mandates.

- **Bias mitigation:** The AI Act explicitly requires manufacturers to identify and correct algorithmic biases, to ensure that AI models do not disadvantage specific patient groups. While the MDR clearly addresses clinical safety and effectiveness from a broader perspective, it does not impose dedicated bias control requirements for AI.

  This means that an AI-based medical device could meet the MDR's clinical validation criteria but fail to meet AI Act compliance requirements due to inadequate bias mitigation. Notified bodies must therefore incorporate bias audits into conformity assessments, to confirm alignment between AI fairness obligations and traditional clinical validation frameworks.

To address these discrepancies, notified bodies must adopt a harmonised assessment approach that integrates AI Act obligations with MDR conformity pathways. Establishing clear criteria for AI model updates, aligning risk classification methodologies, and balancing transparency with intellectual property protection will be essential. Additionally, notified bodies should collaborate with AI regulatory authorities to develop AI-specific conformity guidelines. This will ensure consistency in how AI-powered medical devices are evaluated across both frameworks.

Ensuring compliance with the AI Act and MDR

Simultaneous application of the AI Act and MDR presents unique challenges for the medical device industry. While the AI Act can be considered horizontal legislation, covering many industries and sectors, the MDR is very much vertical legislation.

Some key considerations for medical device manufacturers are highlighted below:

- **Conformity assessment:** Both the AI Act and MDR necessitate third-party conformity assessments, but their criteria and processes differ. This could lead to duplication of efforts and an increased administrative burden. Choosing a notified body with the capabilities and expertise to assess devices according to both the AI Act and MDR should be a priority for manufacturers. Indeed, the AI Act recognises the importance of high-risk AI system assessment (under the AI Act) as being part of the conformity assessment for the MDR, i.e., simultaneously[24].

- **Harmonisation issues:** The AI Act emphasises ethical principles such as transparency and human oversight, while the MDR focuses on clinical safety and performance. Aligning these priorities can be complex and time-consuming.

- **Evolving AI capabilities:** AI systems that learn and adapt over time may require continuous updates to conformity assessments under both the AI Act and MDR. This complicates compliance processes, and it is likely that manufacturers and notified bodies will be challenged by the criteria differences for a significant/substantial change of an AI system.

  For example, as per recital 124 of the AI Act, changes that may affect the compliance of a high-risk AI system with the Act could mean that the AI system should be considered a new AI system that necessitates a new conformity assessment. Examples include a change to the operating system or software architecture, or when there is a change to its intended purpose.

  Although the MDR does not specifically define the criteria for substantial change in the context of AI, this is likely to align with how an evaluation under the MDR would define a substantial change. However, the AI Act goes on to state that changes to algorithms and the performance of AI systems that continue to 'learn', following placement on the market, would not be classified as a substantial change requiring a new conformity assessment, on the condition that such changes were pre-determined by the provider and accounted for during the conformity assessment procedure. This is highly likely to present a challenge to the conformity assessment procedure for both manufacturers and notified bodies. It raises the question: To what extent can changes be "pre-determined" in the ever-evolving landscape of AI, particularly in relation to the clinical setting where interactions between device and patient are rarely uniform?

Manufacturers must implement a robust risk management system that identifies, evaluates and mitigates risks associated with AI components in medical devices. This includes ongoing assessments throughout the life cycle of the AI system, as well as accounting for a degree of foresight as to how the AI system may evolve over time and how the risk-benefit profile of the device may alter accordingly.

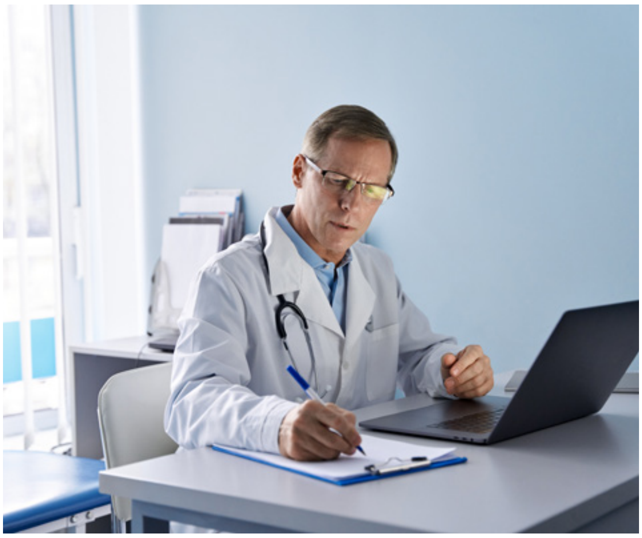
- **Data management requirements (including governance):** While the MDR already mandates robust data management for clinical evaluations, the AI Act adds additional requirement layers related to training data quality, bias mitigation, and algorithmic transparency. Manufacturers must establish clear data governance practices to ensure high-quality training and testing datasets, avoidance of biases in AI algorithms, and protection of patient data in compliance with the EU General Data Protection Regulation 2016/679.

- **Expertise gaps:** Navigating the requirements of both frameworks necessitates multidisciplinary expertise in medical devices, AI technologies, and regulatory compliance. While this may strain resources for smaller manufacturers, they have been considered in the process (see the section on regulatory sandboxes below). Manufacturers must allocate resources to meet the AI Act's requirements, which may include hiring compliance specialists and investing in updated technologies for monitoring and reporting.

- **Cost implications:** Meeting these dual regulatory requirements may increase development and operational costs, potentially impacting the affordability and accessibility of AI-driven medical devices.

The joint Team-NB/IG-NB position paper: AI in medical devices questionnaire

A collaborative group, including TÜV SÜD, IG-NB (German Notified Bodies Alliance for Medical Devices) and Team NB (The European Association of Medical Devices Notified Bodies) has released a questionnaire addressing AI in medical devices. This simplifies the conformity assessment procedure for medical devices that incorporate AI. It is freely available for download and use here: https://www.ig-nb.de/veroeffentlichungen.



The questionnaire provides a solid foundation to understand the relevant standards and state of the art for AI device conformity assessment, as guidance documents and additional clarity from the European Commission becomes available over time. We recommend that manufacturers assess their products using this questionnaire. If any gaps are identified, they should be addressed to ensure a strong foundation for future product development.

While the questionnaire is not currently taking AI Act requirements into consideration, it does allow manufacturers to develop an understanding of how notified bodies are performing conformity assessments for AI devices. This understanding will also benefit manufacturers in the future, as additional requirements from the AI Act are introduced.

Regulatory sandboxes

The AI Act introduces the concept of regulatory sandboxes. The intention is to allow competent authorities to establish controlled frameworks for providers of AI systems to develop, train, validate, and test their innovative AI systems for a limited time under regulatory supervision[25].

The exact terms and conditions of regulatory sandboxes are still to be defined, and it is therefore unclear at present how they will function and operate. However, the AI Act does make clear the timelines for implementation and monitoring of such regulatory sandboxes.

By 2 August 2026 it is expected that Member States' competent authorities will need to have established at least one AI regulatory sandbox at the national level. Annual reporting will also be mandatory, with Member States required to submit annual reports to the AI Office and the Board one year after the sandboxes have been established.

The AI Act also stipulates the need for Member States to facilitate priority access to the AI regulatory sandboxes for small and medium-sized enterprises, including start-ups with a registered office or branch in the EU[26].

Conclusion

The EU AI Act is the first comprehensive legal framework for AI and sets a high bar for its integration in medical devices by emphasising safety, transparency and ethical considerations. While the Act aims to balance innovation support with robust oversight, it does introduce regulatory burdens. This is creating market concern that it could hinder innovation and give the EU a disadvantage in the global environment. Meanwhile, exemptions for lawful medical practices and support for SMEs indicate an attempt to foster innovation alongside compliance.

While there will be compliance challenges, the opportunities for the wider healthcare industry and patients are immense. However, manufacturers, notified bodies and healthcare providers must collaborate early and effectively to ensure the seamless application of this groundbreaking legislation, so that patient outcomes are improved.

Glossary

Glossary of acronyms

GPAI General purpose artificial intelligence
IG-NB The Industry Group of Notified Bodies
MDR Medical Device Directive

Footnotes

1. European Commission Coordinated Plan on Artificial Intelligence https://digital-strategy.ec.europa.eu/en/policies/plan-ai
2. European Commission White Paper on Artificial Intelligence: a European approach to excellence and trust https://commission.europa.eu/publications/white-paper-artificial-intelligence-european-approach-excellence-and-trust_en
3. As per the AI Act, Article 1 (2a) and Article 113
4. Recital 178 of the AI Act
5. Recital 179 of the AI Act
6. Article 2 (1a) of the AI Act
7. Article 2 (1c) of the AI Act
8. Article 2 (1b) of the AI Act
9. Article 99 (3) of the AI Act
10. Recital 29 of the AI Act
11. Recital 29 of the AI Act
12. Article 5 (1f) of the AI Act
13. Article 6 (1) of the AI Act
14. Article 6 (2,3) of the AI Act
15. Article 6 (5) of the AI Act
16. Recital 101 of the AI Act
17. Article 55 (1a-d) of the AI Act
18. Recital 116 of the AI Act
19. Article 50 (1) of the AI Act
20. Article 50 (2) of the AI Act
21. Recital 72 of the AI Act
22. Article 2 (6, 10) of the AI Act
23. Article 2 (12) of the AI Act
24. Recital 124 of the AI Act
25. Article 3 (55) of the AI Act
26. Recital 143 of the AI Act

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